

KnowFlow Core — Technical Architecture & Project Spec

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1. System Overview & Engineering Objectives

KnowFlow AI is a high-throughput, multi-tenant Retrieval-Augmented Generation (RAG) platform. The system delivers sub-200ms Time-To-First-Token (TTFT) across millions of indexed enterprise documents while enforcing strict tenant-level cryptographic data isolation across PostgreSQL, Redis, and vector search indices.

Metric / SLO	Target Threshold	Production Enforcement
P95 Vector Search Latency	< 45ms	PostgreSQL 16 pgvector HNSW (m=16, ef=64)
P99 Streaming TTFT	< 350ms	Multi-Tier Cascading LLM failover protocol
Multi-Tenant Isolation	100% Cryptographic	Workspace UUID partitioning at DB & cache layer
System Availability	99.95% Uptime	Dual-region active-passive worker failover

2. Microservices Topology & Distributed Ingestion Pipeline

The KnowFlow platform decouples interactive query execution from asynchronous document parsing through a resilient Celery worker fleet:

- Ingestion Engine:** Supports PDF (PyMuPDF/Unstructured), Word DOCX, Markdown, and TXT formats with automatic MIME verification.
- Deterministic Chunking:** 500-token sliding window with 100-token overlap, preserving semantic heading hierarchies.
- Vector Embedding:** 768-dimensional dense vector embeddings generated via Google Gemini / Hugging Face models.
- Asynchronous Task Queue:** Redis 7 acts as the broker with exponential backoff retry (max 3 attempts).

3. Multi-Tier LLM Cascading & Automatic Failover

To achieve zero-downtime streaming inference against third-party rate limits (HTTP 429) or upstream outages (HTTP 503), the LLM executor evaluates a three-tier cascade:

- Tier 1 (Primary):** Groq Cloud (Qwen 3.8-27B / Compound-Mini) for ultra-low latency < 200ms TTFT.
- Tier 2 (Secondary):** Google Gemini (Gemini Flash Lite) for complex multi-page synthesis.
- Tier 3 (Tertiary):** Hugging Face Serverless Inference (Llama 3.1 8B Instruct).

4. Security, Cryptography & Role-Based Access Control (RBAC)

KnowFlow implements strict defense-in-depth isolation: workspace memberships are verified via JWT bearer tokens and evaluated against three RBAC roles (ADMIN, MANAGER, EMPLOYEE). Workspace invitation tokens are generated using 32-byte cryptographic random entropy (secrets.token_urlsafes) and stored as SHA-256 hashes.